

Jacob Mattie

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Full-Stack Developer with a mathematics background. A scholar at heart; most motivated by a good question

Languages	Data Pipelines	Backend	DevOps	Infrastructure	
Bash	BeautifulSoup	Celery	CI/CD	Docker	Debian
C++	.csv, .pdf, .xlsx	Celery-Beat	Git	ESP32	Fedora
HTML5 / SCSS	JSON	Django	Debugging	GPIO	SELinux
MATLAB / Octave	PostgreSQL	ETL pipelines	journalctl	systemd	SSH
Python	Redis	Gunicorn	QA/Testing	UART	Ubuntu
SQL	Requests	Nginx	Refactoring	Windows	
	Selenium	REST APIs	rsync		

BSc. Mathematics, 2022, Simon Fraser University

Work Experience

Investment Analyst — Data Platforms & Backend Systems

Aligned Capital Partners / River Wealth

Vancouver, BC

11/2024 - Ongoing

Q: What data exists that might let us predict stock growth and market behaviour?

- Developed production data pipelines supporting investment analysis workflows for a portfolio manager with \$220M+ in assets under management, providing reliable time-series financial data for decision support systems.
- Engineered distributed ETL pipelines using Selenium, Requests, and file-based ingestion, orchestrated via Redis-backed Celery workers, capturing and processing 45,000+ daily financial data points.
- Implemented stateful retry and validation logic across ingestion workflows, ensuring fault tolerance, deduplication, and data consistency under unreliable external data sources.
- Configured a Linux-based server stack (Fedora, Gunicorn, Nginx) to host a Django application, enabling internal access to data analytics and automation tools.
- Designed and operated a CI/CD-like deployment pipeline using Git-based workflows, automated validation tests, and SSH/rsync-based artifact synchronization to a headless Linux environment.
- Automated 20% of company-wide workload through Python scripting and process standardization.

Test Technologist

IBC Technologies

Burnaby, BC

6/2022 – 9/2023

Q: How can we safely optimize heat transfer from a boiler's combustion chamber?

- Automated hardware validation processes by leveraging IoT microcontrollers and PLC systems, streaming test data into cloud-connected repositories for analysis.
- Built high-throughput data analysis workflows in MATLAB for multi-million-point datasets, reducing certification validation time by over 600%.

IT Technician (Internship)

BC Cancer Research Centre (BCCRC)

Vancouver, BC

09/2018 – 01/2019

Q: How much digital freedom can we grant researchers without compromising network security?

- Managed user access controls and permissions within a segregated research network, enforcing data isolation and cybersecurity policies across multiple user groups.
- Assisted in network recovery during facility-wide outage caused by a malware incident, supporting administrators in restoring connectivity and validating system integrity across endpoints.

Volunteering

Automation Technologist

TRIUMF - Canada's Particle Accelerator

Vancouver, British Columbia

05/2025 - Ongoing

Q: What modern technological advances have resolved longstanding engineering problems?

- Contributed the technical foundations of 3 scientific publications (in preparation) on distributed control systems in nuclear physics laboratories.
- Developed a containerized EPICS deployment pipeline (Docker + Bash), leading a modernization initiative for a legacy control software operating over \$1.5B in infrastructure.
- Coded a Raspberry Pi-based stepper motor control system, merging hardware wiring, driver design, and Python-based interface through a transistor array for automated and repeatable experimental control.